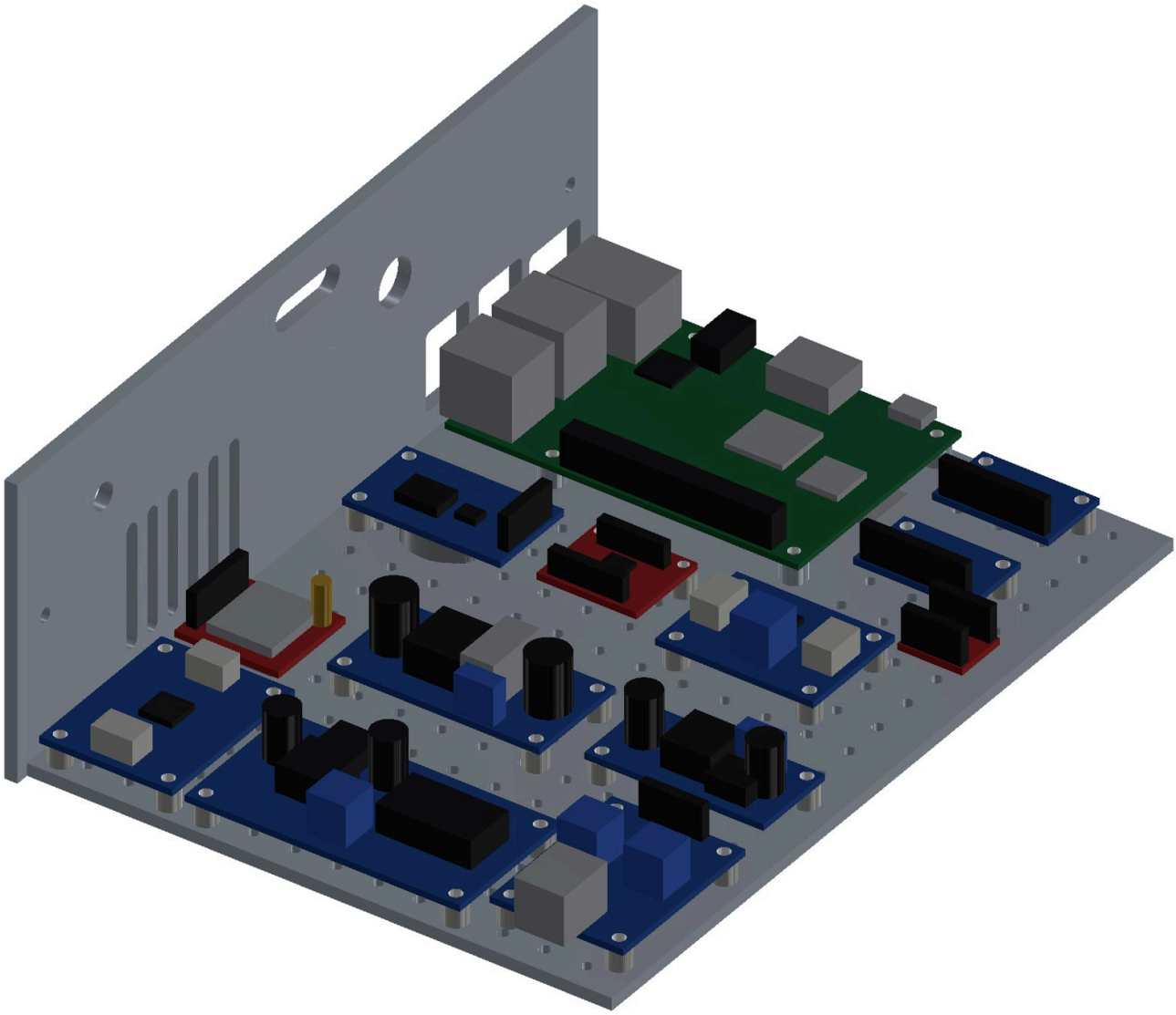
The electronics, in SOLIDWORKS

Our Raspberry Pi electronics as real SOLIDWORKS parts, assembled on the printed tray of the 1:5 live model, checked for clashes, and documented in a SOLIDWORKS drawing with a bill of materials and the full wiring tables.



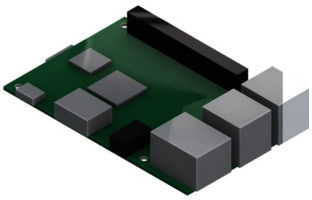
Electronics bay assembly (SOLIDWORKS 2026)

What is in this document

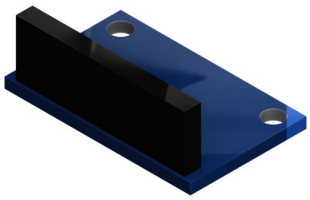
13 native SOLIDWORKS parts · the bay assembly on the LD-S08 tray · interference check and the two fixes it led to · thermal check of the closed plinth · and, at the end, the two A3 sheets of drawing AS-LIVE-ELEC-01 exported from SOLIDWORKS (BOM, balloons, pin map, ADC channels, power rails).

13 parts, modelled in SOLIDWORKS

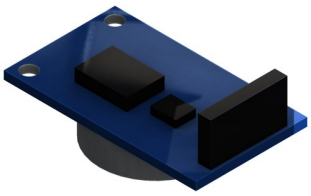
Each part is built from sketches and extrusions, so it has a real feature tree: the board, its mounting holes, then every connector, chip, inductor and capacitor as a named feature (the Raspberry Pi has 16, from the 40-pin header to the micro-SD slot). Board sizes and hole patterns follow the module datasheets and listings (the Pi follows the official mechanical drawing: 85 × 56 mm, M2.5 holes on 58 × 49 mm). Small details are representative.



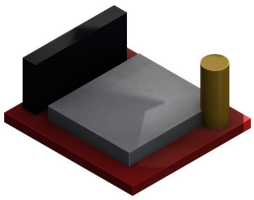
Raspberry Pi 3 Model B+
16 features



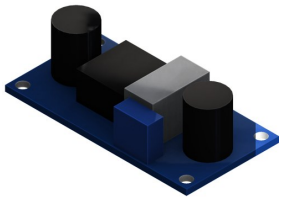
ADS1115 16-bit ADC module
6 features



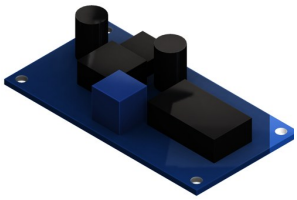
DS3231 real-time clock (ZS-042)
7 features



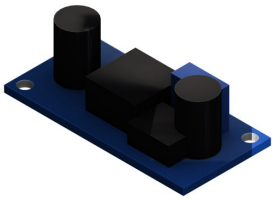
SIM800L GSM/GPRS module
5 features



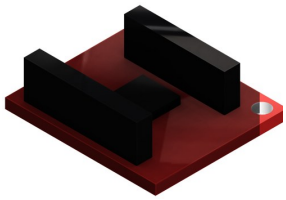
Buck A: 5 V 5 A step-down (Pi supply)
10 features



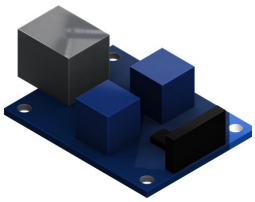
Buck B: LM2596 with voltmeter (SIM800L 4.0 V)
11 features



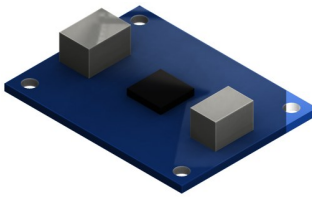
Buck C: LM2596S (5 V sensor rail)
8 features



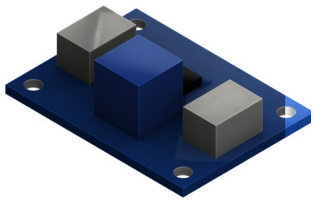
MX1508 "mini L298N" pump driver
6 features



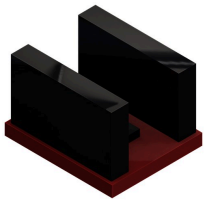
pH board PH-4502C
10 features



TDS Meter V1.0 board
8 features



Turbidity sensor adapter (SEN0189)
9 features



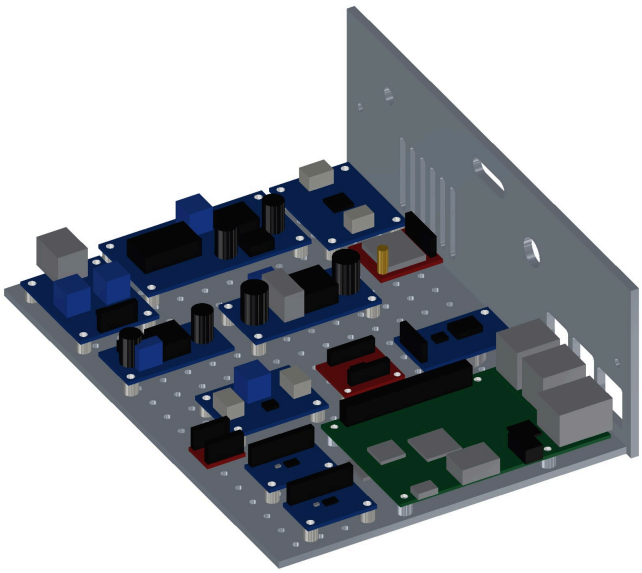
4-channel logic level converter
4 features



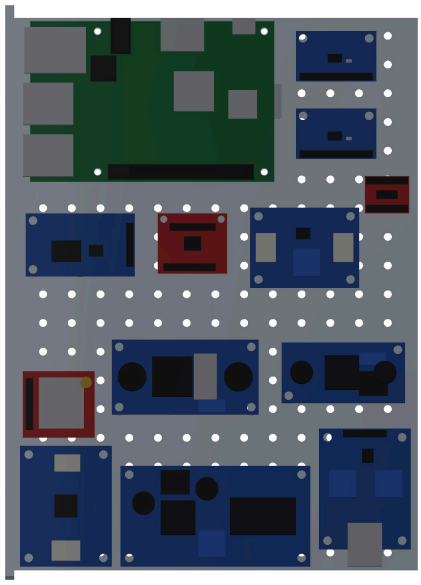
Nylon standoff M3 × 5
2 features

The bay assembly

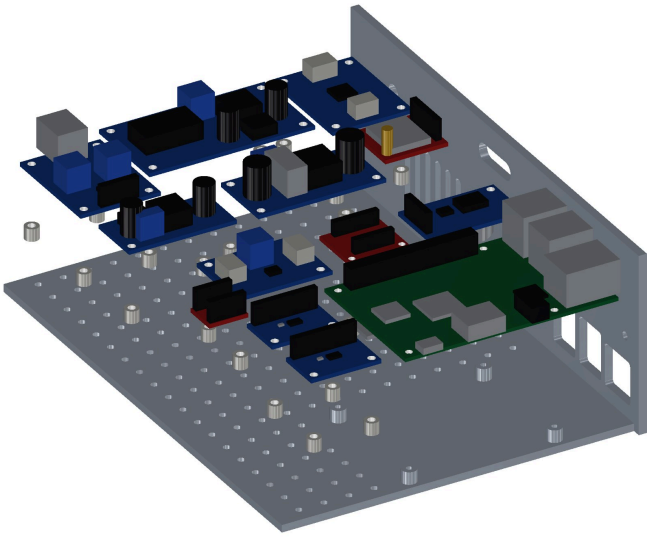
The boards sit on the printed tray LD-S08 that slides into the plinth of the 1:5 model. The Raspberry Pi uses the tray's own four printed standoffs (we measured them on the print file: 58 × 49 mm pattern, tops 5 mm above the floor) with its USB and Ethernet ports facing the end-panel openings. Every other board sits on M3 × 5 nylon standoffs. 44 components in total.



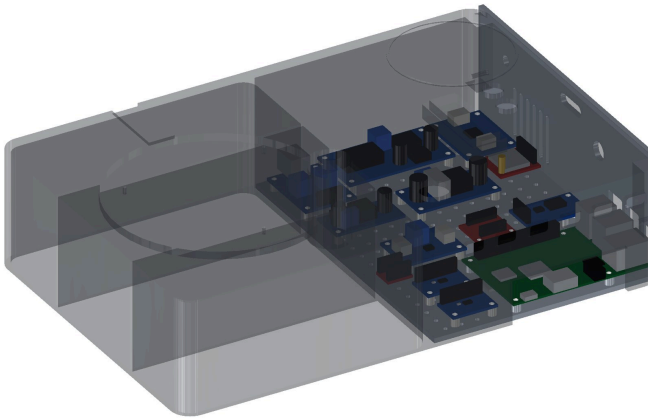
From the open side



Plan view



Boards lifted off their standoffs



Inside the plinth

Design check: interference detection

We ran SOLIDWORKS Interference Detection on the whole bay. The first layout was not right:

Run	Result	What we changed
1	2 interferences: SIM800L × TDS board (114.35 mm^3) and the SIM800L's SIM holder × a standoff (0.57 mm^3)	SIM800L moved 6 mm towards the antenna opening in the end panel
1b	Bounding-box check against the plinth: the pH board's BNC socket stood 1 mm outside the plinth wall (y = −101 mm, wall at −100 mm). Interference Detection cannot see this because that spot is open air in the model	pH board moved 6 mm inwards
2	**0 interferences**, every board inside the plinth	–

RESULT

The final layout fits the printed tray with all 13 boards, 30 standoffs and clear cable space along the centre line.

Board positions (tray frame, mm)

Part	Rotation	x	y
E01_Raspberry_Pi_3B_plus	180°	-144.5	39.0
E02_ADS1115_ADC	0°	-52	74
E02_ADS1115_ADC	0°	-52	47
E12_Level_converter	0°	-28	28
E03_DS3231_RTC	0°	-146	6
E08_MX1508_pump_driver	0°	-100	7
E11_Turbidity_adapter	0°	-68	2
E04_SIM800L_GSM	0°	-147	-50
E05_Buck_A_5V_5A	0°	-116	-42
E07_Buck_C_LM2596S_5V	0°	-57	-38
E06_Buck_B_SIM800L_4V	0°	-113	-95
E09_pH_board_PH4502C	90°	-44	-89
E10_TDS_board_SEN0244	90°	-148	-95

Thermal check of the closed plinth

Will the Raspberry Pi overheat inside the plinth? A first estimate by hand, with the power figures of our electronics design (section 5.3):

Item	Value	Basis
Heat in the bay	3.2 W	Pi 2.5 W typical, sensors 0.15 W, SIM800L idle 0.07 W, 15 % converter loss
Plinth outer area	0.192 m^2	300 × 200 × 72 mm
Overall coefficient	4.0 W/m^2K	outside 10 (convection + radiation), 3 mm PLA (k 0.13), inside 8 W/m^2K: assumed values
Bay air rise	**about 4 K**	Q / (U · A), vents ignored (so on the safe side)
Pi board rise over bay air	about 20 K	2.5 W over about 0.0125 m^2 at 10 W/m^2K
SoC above board average	about 10 K	assumed, typical for the Pi 3B+

Room	SoC estimate	What it means (Pi 3B+ slows its clock above 60 °C, shuts down at 85 °C)
25 °C	about 59 °C	fine; matches what a Pi 3B+ in a case usually shows
30 °C (booth)	about 64 °C	may slow slightly; harmless for logging every 15 min
45 °C (field, always on)	about 79 °C	too close to the limit, so the field unit runs in timed mode: Pi on 3 of every 15 min, 0.5 W average

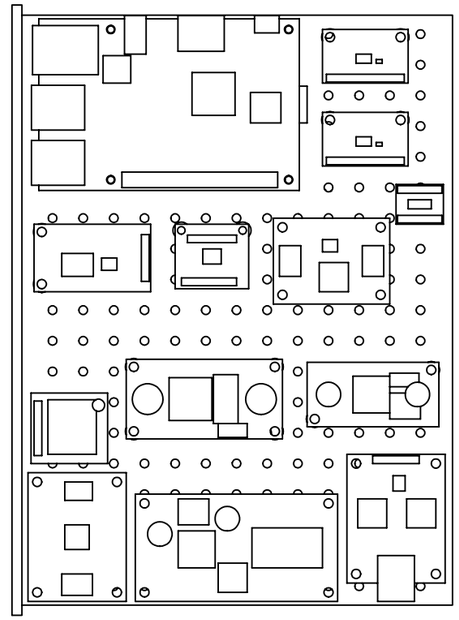
WHAT WE DO ABOUT IT

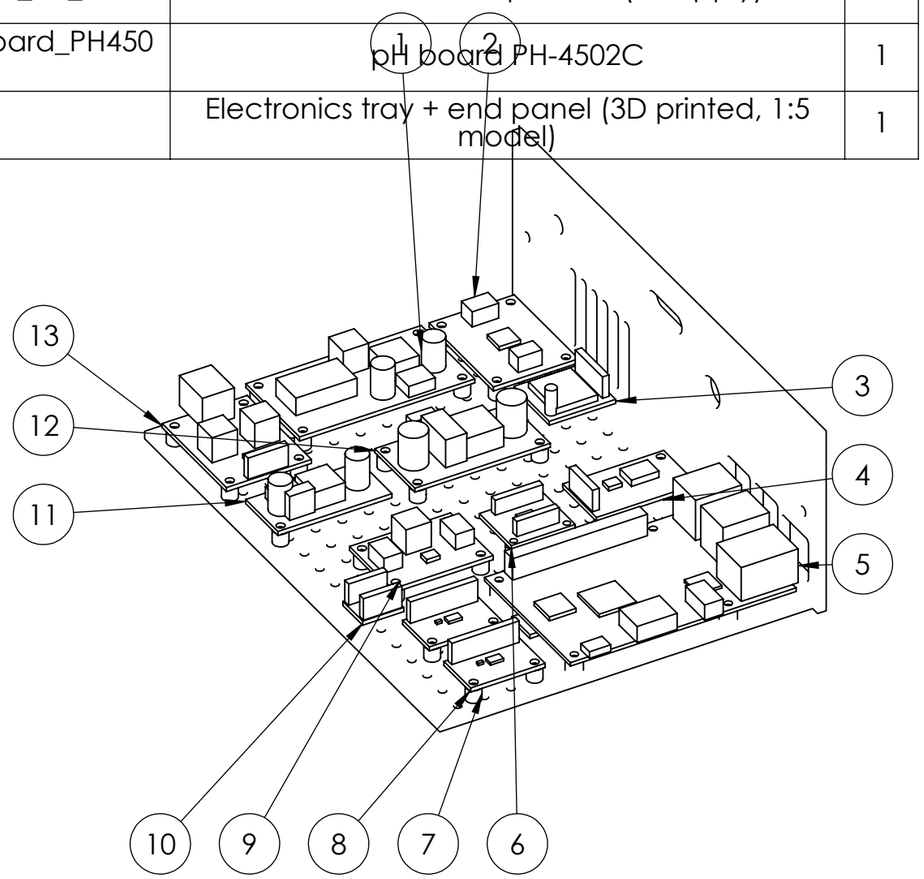
A ₹50 stick-on aluminium heatsink on the SoC, the end-panel vents kept clear, and timed mode in the field.

NEXT: FLOW SIMULATION

This is a hand estimate with assumed film coefficients. We will confirm it with SOLIDWORKS Flow Simulation 2026 (installed): external analysis, gravity −Z, air at 30 °C and 45 °C, heat conduction in solids on, volume heat sources Pi SoC 1.5 W, rest of the Pi board 1.0 W, converters 0.5 W in total, PCB material from the engineering database, goal = maximum SoC temperature.

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	E06_Buck_B_SIM800L_4V	Buck B: LM2596 with voltmeter (SIM800L 4.0 V)	1
2	E10_TDS_board_SEN0244	TDS Meter V1.0 board	1
3	E04_SIM800L_GSM	SIM800L GSM/GPRS module	1
4	E03_DS3231_RTC	DS3231 real-time clock (ZS-042)	1
5	E01_Raspberry_Pi_3B_plus	Raspberry Pi 3 Model B+	1
6	E08_MX1508_pump_driver	MX1508 "mini L298N" pump driver	1
7	E13_Standoff_M3x5_nylon	Nylon standoff M3 × 5	30
8	E02_ADS1115_ADC	ADS1115 16-bit ADC module	2
9	E11_Turbidity_adapter	Turbidity sensor adapter (SEN0189)	1
10	E12_Level_converter	4-channel logic level converter	1
11	E07_Buck_C_LM2596S_5V	Buck C: LM2596S (5 V sensor rail)	1
12	E05_Buck_A_5V_5A	Buck A: 5 V 5 A step-down (Pi supply)	1
13	E09_pH_board_PH4502C	pH board PH-4502C	1
14	LD-S08	Electronics tray + end panel (3D printed, 1:5 model)	1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE			TITLE: Electronics bay, 1:5 LIVE model	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. AS-LIVE-ELEC-01	A3
				WEIGHT:		SCALE:1:2	SHEET 1 OF 2

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RASPBERRY PI 3B+ PIN MAP (BCM)

Pi pin	GPIO	Connects to	Notes
1	3.3 V	ADS1115 x2, DS3231, DHT11, DS18B20 pull-up, rain + LDR modules	about 20 mA total
2, 4	5 V	Pi supply from Buck A	external 3 A fuse
3 / 5	GPIO2 / 3	I2C SDA / SCL: ADS1115 0x48, 0x49, DS3231 0x68	module pull-ups
7	GPIO4	DS18B20 data (1-Wire)	4.7 k to 3.3 V
8	GPIO14 TXD	SIM800L RXD via 1 k / 2 k divider	3.3 V to 2.2 V
10	GPIO15 RXD	SIM800L TXD	2.8 V level, read OK
11	GPIO17	DHT11 data	dtoverlay dht11
12	GPIO18	SIM800L RST	
13	GPIO27	Rain module DO	
15	GPIO22	Rain module power switch	on only while reading
16	GPIO23	MX1508 IN1 (pump)	IN2 tied low
29	GPIO5	LDR module DO	
31	GPIO6	pH board 5 V high-side switch	never with GPIO13
33	GPIO13	TDS board high-side switch	
35	GPIO19	MQ-3 heater MOSFET	about 180 mA
36	GPIO16	ACK / reset button	internal pull-up
37	GPIO26	Status LED / beacon	via transistor
6, 9, 14	GND	Star ground at ADS1115 #1	

ADC CHANNELS

ADC	Ch	Signal	Raw range	Conditioning	At the ADC
#1 0x48	A0	pH Po	0-5 V	12 k / 20 k, 1 k + 100 nF	0-3.1 V
#1	A1	TDS	0-2.3 V	1 k + 100 nF	0-2.3 V
#1	A2	Turbidity	0-4.5 V	12 k / 20 k, 1 k + 100 nF	0-2.8 V
#1	A3	MQ-3 AO	0-5 V	12 k / 20 k, 1 k + 100 nF	0-3.1 V
#2 0x49	A0	LDR AO	0-3.3 V	100 nF	0-3.3 V
#2	A1	Rain AO	0-3.3 V	100 nF	0-3.3 V
#2	A2	Battery voltage	0-14.6 V	100 k / 22 k	0-2.6 V
#2	A3	Spare			

POWER RAILS

Rail	From	Feeds	Protection
12 V in	12 V 3 A adapter (bench) or LiFePO4 12.8 V (field)	Buck A, B, C; pump	3 A fuse, reverse polarity, TVS
5.0 V	Buck A (5 V 5 A)	Raspberry Pi 3B+	external fuse on the 5 V pins
4.0 V	Buck B (LM2596 with voltmeter)	SIM800L VBAT	1000 uF low-ESR + 100 nF + 5.1 V Zener
5.0 V	Buck C (LM2596S)	pH, TDS, turbidity, MQ-3 boards	switched per sensor
3.3 V	Raspberry Pi pin 1	ADCs, RTC, DHT11, DS18B20, rain, LDR	about 20 mA

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE:

Electronics bay, 1:5 LIVE model

DWG NO. AS-LIVE-ELEC-01

A3

SCALE:1:1

SHEET 2 OF 2

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